# Library Management System (Java Web Application)

## 1. Overview

The **library management system** is a web-based application with various features to modernize the practice of borrowing books from a library. EBooks are now the new normal and it's only right that we also have a simple and straightforward system in place for users to interact with those books. The library management system allows users to **register and log-in, view available books, borrow and return books, update personal information and reset their login credentials if they have forgotten them.**

## 2. Tech Stack

- Java (Servlets, JSP)
- Apache Tomcat (Web Server)
- SQL Server (Database)
- JDBC (Database Connectivity)
- CSS (Frontend Pages)
- NetBeans (IDE)

## 3. System Architecture

The system follows a basic 3-tier architecture:

- Presentation Layer: JSP pages (UI for users)
- Application Layer: Java Servlets (Backend logic)
- Data Layer: SQL Server database (Persistent storage)

**Flow:**

User – JSP Page – Servlet – Database – Response – UI update

## 4. Database Design

The database is structured using relational design principles to ensure data integrity and efficient access.

Keys Tables:

Users:

- Stores user account information
- Includes email (unique identifier), password, and profile data

Books:

- Stores book details and availability status

Borrow:

- Tracks borrowing and returning of books
- Includes timestamps for audit purposes

Design Specifications:

- Primary Keys used for unique identification of records
- Foreign keys enforce relationships between users and books
- Varchar used for textual data
- Bit used for Boolean status field (availability)
- Datetime used for borrow/return timestamps

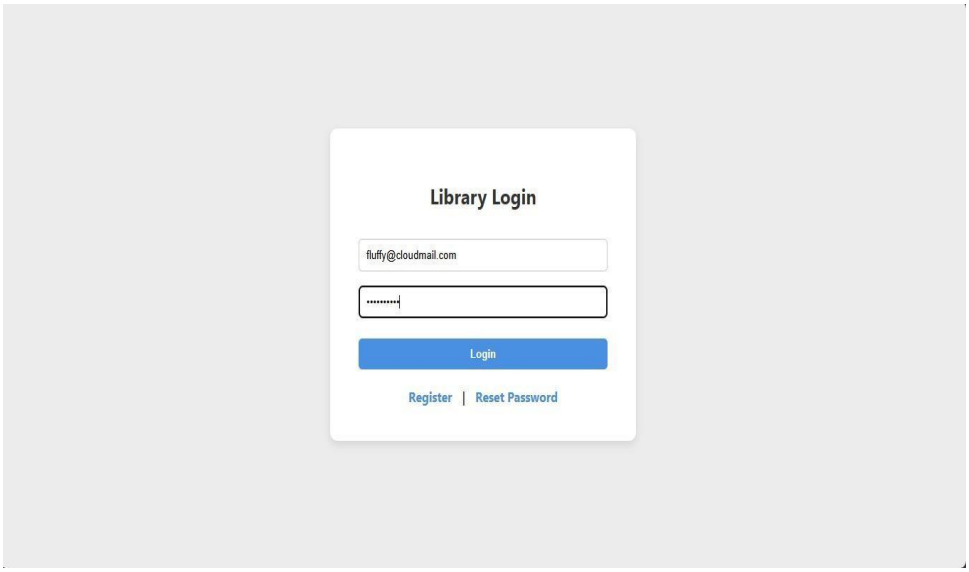
```sql
USE LibraryDB
GO

CREATE TABLE Users (
userID INT PRIMARY KEY IDENTITY,
fullName VARCHAR (255) NOT NULL,
userPassword VARCHAR (255) NOT NULL,
email VARCHAR (255) UNIQUE NOT NULL
)
GO

CREATE TABLE Books (
bookID INT PRIMARY KEY IDENTITY,
Title VARCHAR (255) NOT NULL,
Author VARCHAR (255) NOT NULL,
bookDescription VARCHAR(MAX) NOT NULL,
Avaliable BIT DEFAULT 1
)
GO

CREATE TABLE Borrow (
borrowID INT PRIMARY KEY IDENTITY,
userID INT NOT NULL,
bookID INT NOT NULL,
dateBorrowed DATETIME DEFAULT GETDATE(),
dateReturned DATETIME NULL,
CONSTRAINT userID FOREIGN KEY (userID) REFERENCES Users (userID),
CONSTRAINT bookID FOREIGN KEY (bookID) REFERENCES Books (bookID)
)
GO
```

## 5. Features

- User registration and login system
- Secure authentication and session handling
- View available books
- Borrow and return books
- Profile update functionality
- Password reset system
- Automatic logging of borrow/return actions
- Enforcement of borrowing rules (one active borrow per book)

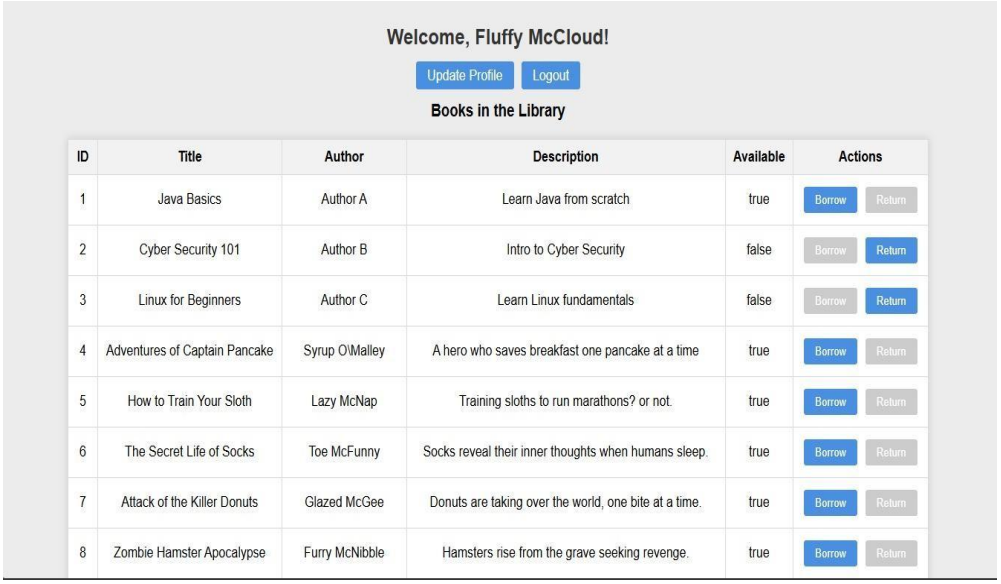
## 6. Navigating to the Library Management System

- Open any web browser of your choice (Edge, Chrome, Firefox, etc.)
- Enter the following URL https://localhost:3128/LibraryManagementSystem/login.jsp
- You will then be directed to the login window as shown in the below screenshot



## 7. Logging in

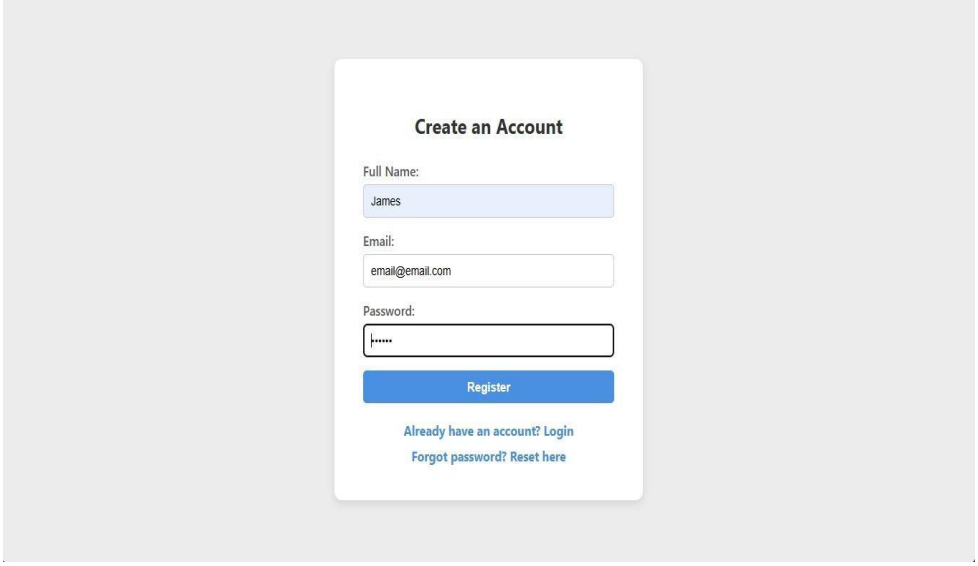
- Enter your registered login credentials (email and password)
- Click **Login**
- You will then be redirected to the dashboard if the login a successful as shown in the below screenshot




| ID | Title | Author | Description | Available | Actions |
|---|---|---|---|---|---|
| 1 | Java Basics | Author A | Learn Java from scratch | true | Borrow Return |
| 2 | Cyber Security 101 | Author B | Intro to Cyber Security | false | Borrow Return |
| 3 | Linux for Beginners | Author C | Learn Linux fundamentals | false | Borrow Return |
| 4 | Adventures of Captain Pancake | Syrup O\Malley | A hero who saves breakfast one pancake at a time | true | Borrow Return |
| 5 | How to Train Your Sloth | Lazy McNap | Training sloths to run marathons? or not. | true | Borrow Return |
| 6 | The Secret Life of Socks | Toe McFunny | Socks reveal their inner thoughts when humans sleep. | true | Borrow Return |
| 7 | Attack of the Killer Donuts | Glazed McGee | Donuts are taking over the world, one bite at a time. | true | Borrow Return |
| 8 | Zombie Hamster Apocalypse | Furry McNibble | Hamsters rise from the grave seeking revenge. | true | Borrow Return |

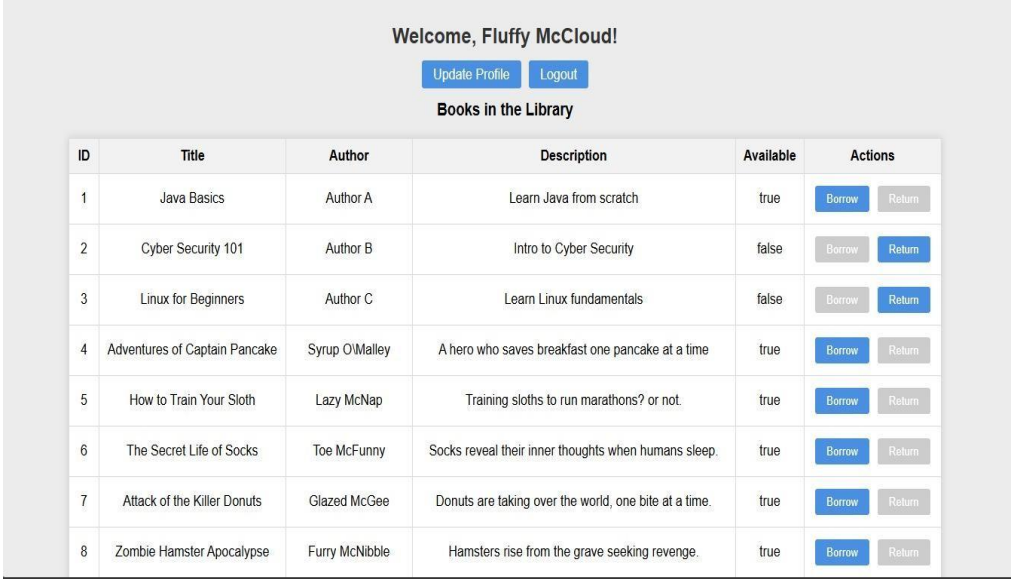
## 8. Registering a new user

- If you do not have any login details, click register on the login window
- Fill in all the required fields with your details (Full Name, Email and Password)
- Click **Register**
- You will then be redirected back to the login window and you can use those details to login.



## 9. Dashboard

- Shows all the details of the available books on the system.
- A user is able to borrow or return a book which is dependent of the availability of the book
- When the book is available the **Borrow** button will illuminate a **blue color** meaning the book is available and can be borrowed
- If the borrow button is **gray** and the **Return button** is blue, it means the book is unavailable and hasn't been return
- Click **Borrow** to borrow a book and **Return** to return a book

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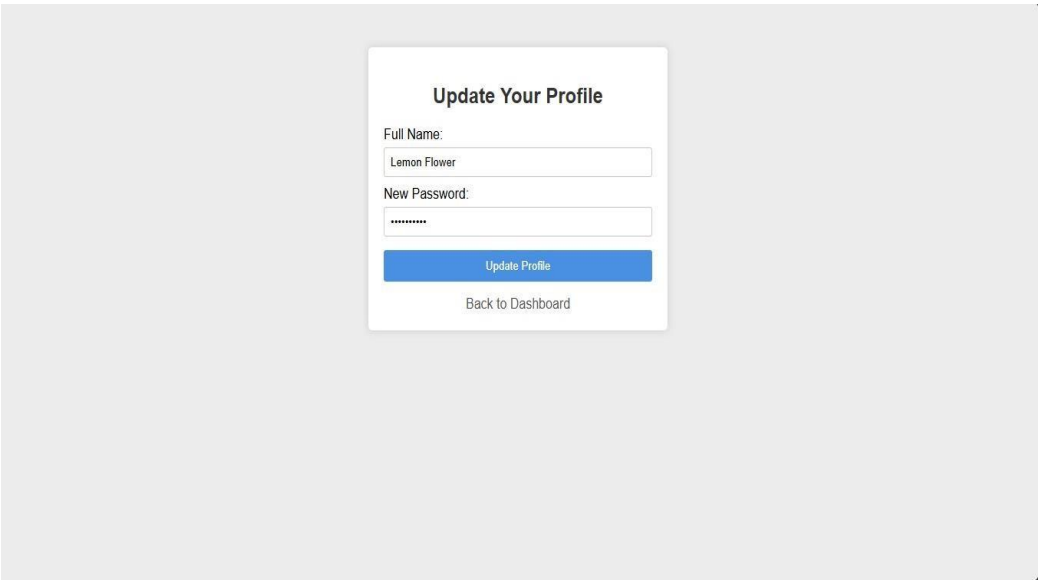
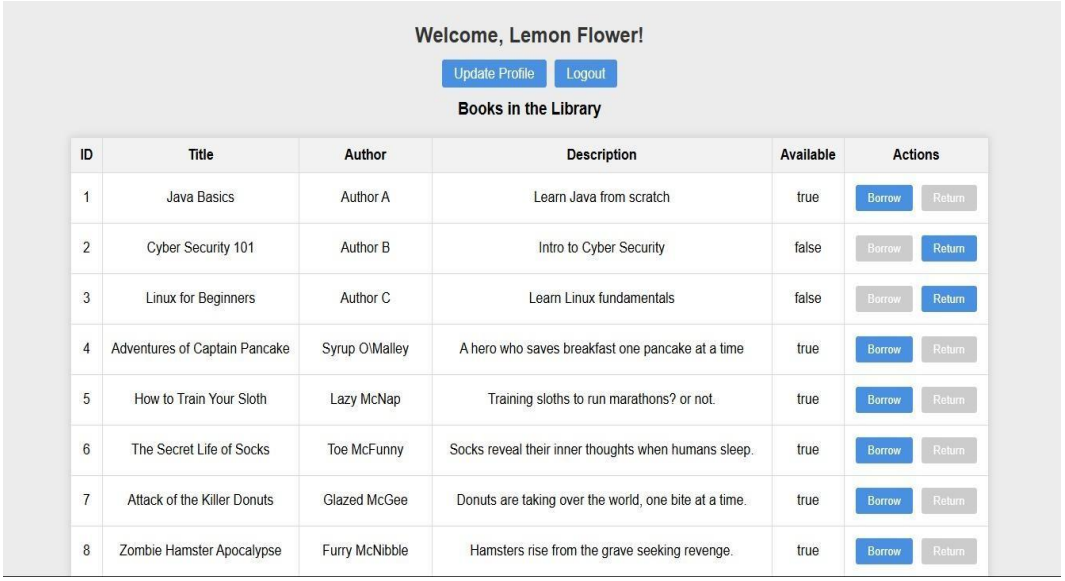
## 10. Update Profile

- If the user wishes to update/changed their name and password, they must click on the **Update Profile** button which will redirect them to the update profile window.
- A user can only update their name and password. The system uses the user registered email to identify the user and update their details.
- Enter the new name you would like to use and the password as well
- If successful, you will then be redirected back to the **dashboard** which will display your name if changed.

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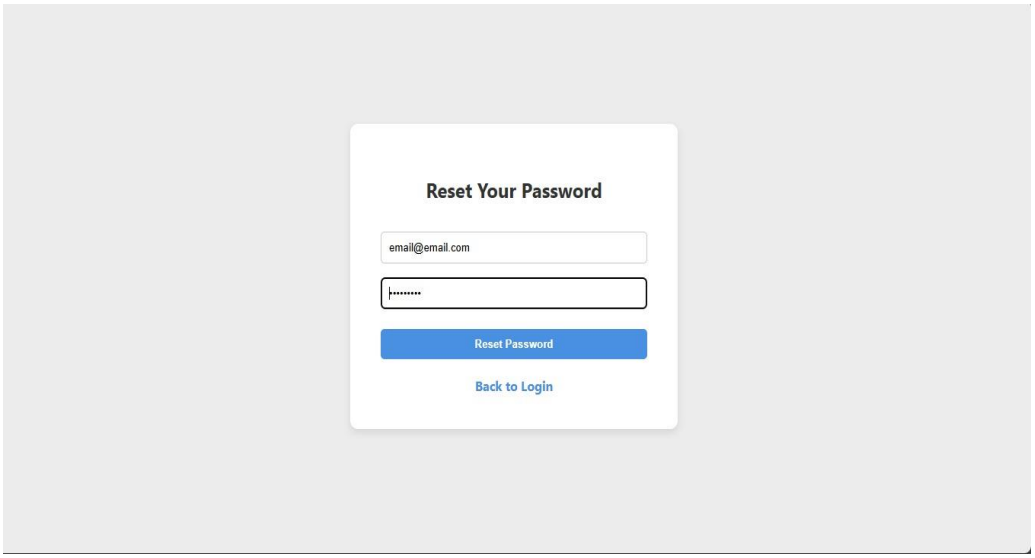
## 11. Reset Password

- Click **Reset Password** on the login window to reset your password.
- Enter your registered email and your new password the click **Reset Password**




## 12. Important Notes

- **Emails** are **unique identifiers** for each registered user
- **Borrowed** and **returned books** are logged automatically into the database on the borrow table.
- **Users cannot** borrow a book that has been borrowed by another user until that user has returned the book
- The system logs the current **exact date and time** that the book was borrowed or returned.